

Noticing language: What echolocation tells us about language emergence

Brett Reynolds

Department of English, Humber Polytechnic & University of Toronto

brett.reynolds@humber.ca*

July 2025

Abstract

Why do humans alone develop language? Rather than specialized processing modules, I argue that evolution selected for a simple attentional bias, hyperpriors that amplify communicative signals (eye contact, infant-directed prosody, hierarchical rhythm). This precision-weighting ensures linguistic patterns dominate learning without requiring language-specific architecture. Echolocation demonstrates the principle. Blind humans achieve remarkable spatial discrimination through self-generated sounds, a complex capacity that emerges without specialized adaptations. This spandrel falsifies the common inference that complex abilities require direct selection. If echolocation can arise from generic neural systems plus experience, so can language, given appropriate attentional focus. The attention hypothesis explains why language emerges universally while echolocation remains rare (only language has inherited attentional support), why other vocal learners never develop language (they lack the relevant hyperpriors), and why adults struggle with low-salience L2 features (they lack children’s automatic orientation to these patterns). Three experiments could test this: cross-species neonatal preference tests, computational models varying only attention parameters, and fNIRS measurement of precision-weighting in newborns. The account reframes language evolution as selection for what we notice, not what we compute.

Keywords: language emergence, attention bias, echolocation, spandrel, gene–culture coevolution, hyperprior

*Working paper | Conceptual synthesis. No new empirical results. Comments welcome. Early ideas appeared in Reynolds (2020). I am grateful to Lore Thaler for detailed comments on that draft and to Randy Harris and Peter Evans for comments on the revision. I used ChatGPT o3 (OpenAI, Apr 2025) and Claude 3.7/4 (Anthropic, May 2025) in drafting and editing. I take full responsibility for the final text and conclusions.

1 Introduction

Why do humans alone develop language? The question has driven decades of debate between two camps. One posits specialized neural architecture that evolved for language (Berwick and Chomsky, 2016; Chomsky, 1965; Hauser, Chomsky, and Fitch, 2002; Fitch, Hauser, and Chomsky, 2005). The other attributes acquisition to domain-general mechanisms shaped by culture (Christiansen and Chater, 2015; Evans and Levinson, 2009; Scholz et al., 2025). Proponents of the first view invoke computational-complexity and poverty-of-stimulus (PoS) arguments, concluding that humans have to possess innate, language-specific modules (Baker, 1979; Chomsky, 1980). But child-scale sequence models already distinguish licit from illicit grammars, sharply undercutting PoS claims (Perfors, Tenenbaum, and Regier, 2011; Warstadt, Mueller, et al., 2023; Kallini et al., 2024). Cultural-evolution accounts, for their part, show how languages adapt to cognitive constraints through transmission (Kirby, Cornish, and Smith, 2008), which in turn bootstraps acquisition (Thompson, Kirby, and Smith, 2016). Still, neither camp explains the deeper puzzle: why humans tune in while every other species tunes out (cf. Scott-Phillips, 2017).

I propose the *attention-bias hypothesis*: humans inherit hyperpriors that boost precision on ostensive, hierarchically rhythmic signals (e.g. syllable-timed prosody). By boosting precision on those cues, the bias steers domain-general learning toward language without specialized computation, rich innate knowledge, or elaborate evolutionary hurdles; a simple gain tweak suffices.

Converging evidence challenges dedicated-faculty accounts. Domain-general action-sequence mechanisms plausibly support language (Kolodny and Edelman, 2018); generic learners in computational models master human languages but stumble on unattested ones (Kallini et al., 2024; Warstadt, Parrish, et al., 2020). Such results suggest language emerges as a SPANDREL – a complex ability arising from general capacities rather than direct selection (Stephen J. Gould and Lewontin, 1979).

The spandrel view gains unexpected support from human echolocation. Specialized-faculty advocates often invoke bat echolocation: bats genetically evolved for ultrasonic navigation, so humans must have done the same for language by the same logic. The analogy collapses once we see that humans echolocate too. Blind individuals, despite no specialized adaptations, navigate with self-generated clicks and achieve spatial resolutions rivaling bats (Teng and Whitney, 2011; Dodsworth, L. J. Norman, and Thaler, 2020). Complexity therefore need not imply genetic adaptation. Crucially, because humans lack an innate bias toward echo patterns, echolocation remains latent in nearly all sighted individuals, underscoring attention’s central role.

Language and echolocation are both complex achievements attainable with domain-general machinery. Evolution tipped the scales only for language, endowing infants with an attentional bias toward communicative signals. We all acquire language because we can’t avoid noticing it; we rarely echolocate because nothing compels us to attend.

This paper develops the attention-bias account in five stages. Section 2 establishes human echolocation as a clear spandrel, undermining arguments for specialized language faculties. Section 3 gathers evidence for innate attentional biases and formalises precision-weighting as the mechanism. Section 4 outlines three empirical tests: cross-species preference studies, computational models that vary only attention parameters, and neonatal neuroimaging of precision-weighting. Section 5 draws implications for language emergence within gene–culture coevolution, and Section 6 closes. The account reframes language evolution as selection for what we notice, not how we compute.

2 Echolocation: Spandrels, Adaptations, and Evolutionary Insight

2.1 Key evolutionary terms

Following Gould and colleagues, I treat three labels as analytically distinct. An ADAPTATION is a trait moulded by natural selection for the function it now serves; the claim requires evidence that historical selection targeted that function and that the trait’s structure shows design for it (Stephen Jay Gould and Vrba, 1982). An EXAPTATION is a trait that performs a current role but originated either under selection for a different role or as a neutral architectural by-product. A SPANDREL is the limiting case of exaptation, a feature that arises unavoidably from the geometry of adjoining adaptations and was never itself a target of selection (Stephen J. Gould and Lewontin, 1979).

Diagnosing which category applies demands converging evidence. Specialized peripheral anatomy argues for adaptation; generic hardware that is merely pressed into new service supports an exaptation or spandrel reading. Dedicated neural circuitry also counts as structural specialization, even in the absence of distinctive anatomy, whereas flexible computational implementation suggests exaptation. Developmental data provide a further clue: a capacity that emerges reliably without tuition is more plausibly an adaptation, while one that appears only with explicit training is better viewed as a spandrel.

2.2 Echolocation across species

Echolocation ranges from rudimentary obstacle detection to millimetre-scale prey tracking. In microchiropteran bats and odontocetes, the capacity is unmistakably an adaptation. Both groups display peripheral specializations: enlarged cochleae, stiffened basilar membranes, and the convergently evolved motor protein *prestin* in outer hair cells (Vater, Meng, and Fox, 2004; Dawkins, 2024). Bat pups begin emitting task-ready pulse trains on their first flights (19–20 d post-natal) without tutelage (Buchler, 1980), and adults resolve prey size, texture, and movement with striking precision (Wilkinson, 2003). Neural pathways from inferior colliculus to auditory cortex are dedicated to echo delay and Doppler

analysis, yielding real-time spatial maps (Aihara, Fujioka, and Hiryu, 2013). Together these anatomical, developmental, and functional lines leave little doubt that ultrasonic echolocation is an adaptation.¹

Humans present a contrasting picture. The auditory periphery is ordinary, operating in the 2–10 kHz band; no structural tweaks hint at selection for echo processing. Nonetheless, both blind and sighted adults can be trained to extract detailed spatial information from mouth-click echoes (Kolarik et al., 2014; Dodsworth, L. J. Norman, and Thaler, 2020). Expert blind echolocators discriminate horizontal offsets of about 1.2° , approaching the limits of human binaural localization and comparable to bat performance on matched tasks (Teng and Whitney, 2011, p. 483). Surveys put uptake among congenitally blind adults near 20–30 %, likely conservative because many users employ the skill tacitly (Thaler, 2013). Latent motor abilities of this sort aren’t unusual: most people neglect the auricular muscles despite having the hardware (Bérzin and Fortinguerra, 1993).

The neural implementation diverges sharply from the bat template. Functional imaging shows that blind echolocators commandeer occipital cortex. Primary visual areas develop coarse retinotopic-like maps for echo delay (Liam J. Norman and Thaler, 2019), the occipital place area tracks echo-defined routes (Liam J. Norman and Thaler, 2023), and parahippocampal cortex codes surface material from echoes (Milne et al., 2015). Short-term training in sighted adults induces parallel shifts, confirming that these changes are experience driven rather than hard-wired (Liam J. Norman, Hartley, and Thaler, 2024). The absence of peripheral specialization, the need for explicit practice, and the repurposing of visual regions together identify human echolocation as a spandrel, not an adaptation.

Cross-modal plasticity shows that human echolocation relies on repurposing existing neural systems rather than dedicated circuits evolved specifically for echolocation. Table 1 summarizes the key neuro-imaging evidence.

2.3 Human echolocation as a spandrel

Because most sighted humans never need, notice, or practise echolocation, the trait can’t have delivered past fitness payoffs and therefore couldn’t have been a direct target of selection. Its hallmarks are those of a spandrel: no peripheral specialization, reliance on neural circuits built for other tasks, and emergence only under explicit training or sensory loss.

The cortical machinery comes largely from vision. Echo cues recruit early visual areas, the occipital place area, and parahippocampal regions, territory originally tuned for spatial layout and surface properties. The auditory periphery remains generic, and behavioural onset is experience dependent, in sharp

¹Putnam’s 1980, p. 405 claim that “*seven-eighths of the bat brain*” is auditory is off by an order of magnitude: in insectivorous species auditory nuclei occupy only 12–46 % of hippocampal volume, within the human range (Hutcheon, Kirsch, and Garland, 2002; Rademacher et al., 2001; Fortin et al., 2008; Rombaux et al., 2010).

Table 1: Neural correlates of human echolocation

Study	Participants	Key finding	Effect metric
Norman & Thaler (2019)	5 blind experts; 10 sighted controls	Retinotopic-like spatial maps for echoes in primary visual cortex	$t_{10} = 3.675$; $d \approx 2.0$ (95 % CI 0.7–3.3) ^a
Norman & Thaler (2023)	6 blind experts; 12 blind controls; 14 sighted controls	Spatial-route echoes elicit stronger occipital-place-area activation than scrambled echoes	Experts vs. sighted: $d \approx 1.5$ ^b
Norman <i>et al.</i> (2024)	26 adults (blind + sighted) pre/post training	Echo training reshapes left-hemisphere V1 representation	Within-subject $F_{1,24} = 4.948$; $d_z \approx 0.44$ ^c
Milne <i>et al.</i> (2015)	3 blind experts; 6 blind controls	Parahippocampal cortex tracks echo-based material judgements	Behavioural 4-AFC gap $d \approx 3.4$ ^d

^a $d = t\sqrt{1/n_1 + 1/n_2}$; CI via Hedges’ g .

^b From published means and pooled SD.

^c $d_z = t/\sqrt{N}$.

^d Imaging single-subject; behavioural statistic shown.

contrast to the automatic, pup-onset system of bats. Human echolocation is neither rudimentary nor ersatz; its performance ceiling rivals bat localization once practice drives cortical re-allocation. What differs is evolutionary provenance, not computational sophistication.

Latent audible-frequency echolocation also appears in shrews and rats (E. Gould, Negus, and Novick, 1964; Rosenzweig, Riley, and Krech, 1955), reinforcing the point that echo processing can hitch-hike on general auditory and spatial circuits. Humans show a parallel dormant channel in tetrachromacy: roughly 12 % of women carry a fourth cone pigment but seldom experience novel colour categories (Jordan *et al.*, 2010). Both cases illustrate how complex capacities can sit unused until circumstances or training expose them.

Human echolocation thus exemplifies a broader principle: plastic neural architecture can yield high-level abilities without prior adaptive design. The capacity is real, potent, and biologically cheap, precisely the profile expected of a spandrel.

2.4 Implications for language-emergence arguments

Adaptationist defences of a language-specific instinct often lean on an analogy to bat echolocation (Pinker and Bloom, 1990; Jackendoff, 2002; Pinker and Jackendoff, 2005; Tooby, Cosmides, and Barrett, 2005; Fitch, 2014). The argument may be spelled out as:

- Premise 1: Bat ultrasonic echolocation is a complex, species-typical capacity.
- Premise 2: It arose through direct selection for that function.
- Premise 3: Human language is likewise complex and species-typical.

- Bridge: Any complex, species-typical capacity has to arise through direct selection for its current role.
- Conclusion: Therefore language is a direct adaptation.

Latent human echolocation breaks the bridge. The ability is complex and widely attainable, but it (i) typically requires deliberate training, (ii) uses unspecialized auditory hardware, and (iii) shows no phylogenetic trace of selection for echo-navigation. By Gouldian criteria it is a spandrel and a counter-example to the inference “complex $\rightarrow$ adaptation.” The echolocation template therefore offers no evidential support for a language-specific module; such a claim must stand on genetic, anatomical, or fitness data of its own. The live evolutionary question becomes: which inherited biases draw human infants – but not other species – to structured vocal-gestural input (Vernes, 2017)?

2.5 Addressing the universality asymmetry

A likely objection is the distributional gap: language emerges in every neurotypical child without instruction, whereas echolocation surfaces only in a minority of blind adults. If both capacities are spandrels, why the asymmetry?

The *attention-bias hypothesis* offers a direct answer: selection, I contend, targeted not the evolution of specialized language processors, but the tuning of an inherited gain mechanism that assigns high precision to ostensive, hierarchically rhythmic signals. That boost magnifies linguistic prediction errors, making them the most informative cues for a generic learner (see Section 3). No comparable bias exists for click echoes, so the latent skill remains dormant unless blindness or formal training forces attention onto those cues.

A plausible evolutionary route is self-domestication: selection against reactive aggression induced neural-crest changes that bias attention toward social signals (Wilkins, Wrangham, and Fitch, 2014; Wrangham and Hare, 2019). Increased *BAZ1B* dosage, for example, shifts neural-crest migration and, in model systems, raises early oxytocin signalling, sharpening precision on eye contact and infant-directed prosody (Zanella et al., 2019; Rokicki et al., 2022). The same circuitry offers no boost for echo acoustics.

Without the attentional gain boost, echolocation stays rare; with it, language becomes universal. The contrast dissolves the “universality asymmetry” while preserving the lesson of the analogy: complex skills can crystallize from domain-general neural stock. The circuitry is generic; only language inherits an orienting mechanism that guarantees uptake.

2.6 Anatomical considerations

Three anatomical traits are routinely cited as evidence of a language-specific adaptation: (i) the descended larynx, (ii) fine orofacial motor control, and (iii) cortical hemispheric lateralization for speech. Each, on closer inspection, is better read as an exaptation with its own adaptive history.

The descended larynx, long championed as a speech device, most plausibly evolved for acoustic size exaggeration in male competition, as indicated by sexual dimorphism and parallel descent in several loud-calling mammals (Fitch and Reby, 2001). Laryngeal descent occurs across multiple lineages during vocal display, so the trait is neither uniquely human nor necessarily language-driven (Fitch, 2010). For an opposing view see Lieberman (2017); Boë et al. (2019) provide a reply.

Primates possessed refined orofacial motor control well before language, primarily for feeding and facial expression (Burrows, 2008). Human elaboration likely reflects further selection for facial signalling and diet; articulate speech then co-opted this pre-existing control.

Hemispheric asymmetry is likewise not diagnostic of language. Functional lateralisation for vocalisation appears widely across vertebrates (Ocklenburg, Ströckens, and Güntürkün, 2013). Songbirds and other vocal-learning taxa show left-dominant processing of species calls, confirming that lateralisation is neither uniquely linguistic nor uniquely human.

These structures therefore supply physical *scaffolding*. Language arises only when that scaffold is combined with three behavioural ingredients: vocal-production learning, ostensive individually addressed signalling, and sensitivity to hierarchical rhythm. To date, humans are the lone species known to express all three simultaneously (Table 2).²

In sum, the descended larynx, orofacial dexterity, and hemispheric specialisation each originated under pressures unrelated to speech and were later recruited for linguistic ends. Their convergence supports the view of language as an emergent capacity built on recycled anatomy rather than as a product of bespoke biological engineering.

2.7 Latent capacity or faint adaptation?

To count human echolocation as a spandrel rather than an under-expressed adaptation we need evidence on five fronts: developmental onset, peripheral anatomy, neural circuitry, cross-modal plasticity, and historical utility. The capacity remains dormant in sighted individuals and appears only after blindness or deliberate training; no peripheral structures differ from the generic human ear; neuro-imaging shows that echo processing co-opts cortical regions built for vision and reallocates them flexibly across individuals; and neither archaeological nor ethnographic records hint at any past reproductive advantage. Together these points mark echolocation as a structural by-product of other selected systems, not a trait shaped by selection in its own right.

²Fitch (2010, Ch. 14) proposes a melodic-rhythmic bias; the present account attributes the critical step to an attentional hyperprior that amplifies *any* hierarchically rhythmic, ostensive stream.

Table 2: Comparative profile of vocal-learning taxa on three properties relevant to the attention-bias hypothesis

Taxon (ex-emplar)	Vocal-production learning	Ostensive individually addressed signals	Sensitivity to hierarchical rhythm / beat
<i>Homo sapiens</i>	Species-universal VPL (Kuhl, 2004)	Gaze + prosody in IDS (Vouloumanos and Werker, 2007)	Spontaneous beat entrainment (Large and Palmer, 2002)
Songbirds (zebra finch)	Juvenile males copy tutor song (Immelmann, 1969)	Directed courtship song; amplitude scales with distance (Brumm and P. J. Slater, 2006)	Discriminate isochronous vs. irregular; poor tempo generalisation (van der Aa, Honing, and ten Cate, 2015)
Parrots (African grey)	>100 referential labels learned (Pepperberg, 1999)	Uses labels to address humans (Pepperberg, 1999)	Cross-tempo beat entrainment (Snowball) (Patel et al., 2009)
Bats (<i>Saccolaryx</i>)	Juveniles learn hover-song patterns (Knörnschild et al., 2010)	Hover-song directed at one harem female (Behr and Helversen, 2004)	No beat entrainment reported
Cetaceans (bottlenose dolphin)	Calves invent signature whistles; adults copy others’ whistles (Janik and P. J. B. Slater, 1998)	Whistle copying addresses a specific individual (King and Janik, 2013)	Turn-taking 1 s (Nakahara and Miyazaki, 2011); no beat entrainment
Pinnipeds (California sea lion)	Contextual vocal imitation	–	Head-bob entrainment to musical beats

3 The Attention Hypothesis

3.1 Attention as biological gain control

Humans come equipped with a neural “volume knob” that amplifies signals likely to be communicative. In predictive-processing terms the knob sets the PRECISION of prediction errors: high precision forces model revision, low precision lets mismatches pass as noise.³

Two cross-modal signatures trigger the boost in infants: (i) *ostensive cues*, such as eye contact, infant-directed prosody or motion – and (ii) *hierarchical rhythm* whose envelope chunks into syllable- or gesture-sized units. When both co-occur the stream is tagged “communicative” and routed through a high-gain channel; lacking either cue, it stays in the low-gain channel.

Four-step loop.

1. Detect: Subcortical circuits spot an ostensive cue and tag the input as communicative.
2. Boost: Tagging raises precision across the auditory (or visual) hierarchy.
3. Update: Amplified prediction errors drive rapid model adjustment.

³A formal derivation of the precision-weighting rule appears in the supplementary materials.

4. Stabilise: Synaptic weights consolidate the pattern; similar structure in low-gain noise is ignored.

The outcome is that the single inherited rule, *treat ostensive rhythmic streams as reliable*, steers a generic learner toward the linguistic patterns of its community. No dedicated syntax module is required; attention-driven gain control does the heavy lifting.

3.2 Distinguishing the attention-bias hypothesis

Usage-based accounts and Schmidt’s *Noticing* hypothesis assign the bottleneck to conscious awareness or to the learner’s pattern-finding machinery (Schmidt, 1990). Tomasello’s social-cognitive model adds intention-reading as a second prerequisite (Tomasello, 2003). By contrast, the attention-bias proposal places the genetic change one level earlier: in a set of hyperpriors that up-weight ostensive, hierarchically rhythmic input. The proposal keeps both pattern-finding and theory-of-mind domain-general; what is inherited is a precision filter that decides which data reach those mechanisms with full weight.

Deacon’s account of the symbolic leap (Deacon, 1997) requires sustained focus on linguistic form rather than on literal reference. A built-in precision boost for communicative streams supplies that focus automatically, accomplishing the same explanatory work without positing new analytic machinery.

Laboratory work supports the distinction. Infants segment artificial grammars from non-social audio, yet retention and generalization rise sharply when the same material is framed ostensively (Saffran, Aslin, and Newport, 1996; Kuhl, Tsao, and Liu, 2003). Social context therefore seems to raise precision on the relevant prediction errors.

One might suggest that caregiver behaviour alone explains human language acquisition: in naturalistic field data, infants hear orders of magnitude more infant-directed vocalizations than any ape peer – roughly 70–400× depending on the species (Wegdell et al., 2025). But disparity of input only restates the puzzle: what disposes human caregivers to supply, and human infants to draw, such intensive directed speech? An attention-bias account treats the caregiver side as responsive to infants’ heightened uptake of ostensive cues, making abundant input necessary but not sufficient. Without the perceptual machinery to extract hierarchical patterns, even saturation training fails to yield language-like behaviour, as decades of ape tutoring attest (Terrace, 1979).

Following Scott-Phillips (2015), I use OSTENSIVE CUE to mean any signal explicitly directed at the learner; here, though, ostension functions as an attentional prior, not as a full-blown pragmatic inference.

3.3 Evidence for innate attentional biases

Comparative work shows that many species enter the world with built-in propensities to notice some stimuli over others. These biases channel information uptake and differ not only across taxa but even across strains within a species.

Field-cricket females, for example, orient toward a narrow frequency band that matches the conspecific calling song (~ 4.8 kHz); the ascending interneuron AN1 is tuned to the same band, confirming a hard-wired mate-recognition filter (Wohlers and Huber, 1982). Domesticated foxes, selectively bred for sociability, detect human gaze direction far better than wild conspecifics (Hare et al., 2005). Snake-naïve Japanese macaques spot snake images faster than flower images, demonstrating an inherited bias toward ancestral threats (Shibasaka and Kawai, 2009).

White-crowned sparrow forebrain neurons start broadly tuned but shift their best frequency to the bird’s own emerging song during early practice, a precision boost on the “self-produced motif” channel without peripheral change (Volman, 1993). Social context matters: juvenile zebra finches ignore tape playback but copy the same song when a live tutor is present, an effect interpreted as heightened attentional arousal rather than improved sensory fidelity (Doupe and Kuhl, 1999).

Bank-swallow parents accept any chick until day 15–16; once chicks develop individual “signature calls,” parents learn them rapidly and eject intruders, a gain switch on caller identity (M. D. Beecher, I. M. Beecher, and Hahn, 1981). Such cases illustrate Seligman’s *preparedness*: some cue–outcome pairings are pre-weighted for rapid learning (Seligman, 1970).

Human newborns already prefer speech over acoustically matched non-speech (Vouloumanos and Werker, 2007). By eight months they segment words using transitional probabilities (Saffran, Aslin, and Newport, 1996), yet between six and ten months their discrimination of non-native phonemes declines, indicating a developmental re-weighting parallel to the animal data (Werker and Tees, 1984). Newborns also orient to face-like patterns with “eyes” above a “mouth” even under tight visual control, signalling high initial precision for conspecific faces and gaze (Johnson et al., 1991; Farroni et al., 2005).

These findings show that evolution tunes both the magnitude and the timing of attentional gain without altering basic sensory hardware. The same logic underpins the present hyperprior account: humans inherit high initial precision for ostensive, hierarchically rhythmic signals, ensuring that communicative input carries maximal weight in early learning.

3.4 Attention versus processing accounts

Standard acquisition theories posit specialized machinery (Merge, constraint templates, parameter setting) to bootstrap grammar from scant input. The PoS argument makes that machinery indispensable. The attention-bias view shifts the locus of innateness: selection shaped a gain function that assigns high precision to ostensive, hierarchically rhythmic streams. Amplified prediction errors then drive a generic learner toward linguistic structure without language-specific algorithms.

Apparent support for processing solutions often dissolves under an attentional lens. Mismatch responses to native–non-native contrasts in seven-month-olds

(Friederici, 2005) need not imply built-in phonological categories; infants could reach that discrimination via months of precision-boosted exposure. Likewise, laboratory segmentation of non-social sound strings is fragile, but performance soars when identical material is presented ostensively (Saffran, Aslin, and Newport, 1996; Kuhl, Tsao, and Liu, 2003).

Second-language evidence fits the same pattern. Adult L2 learners master high-salience vocabulary but struggle with low-salience morphology, category systems, and phonetic timing (Ellis, 2006; Schmidt, 1990). An early bias raises precision on those features for infants; adults gave to re-engage it explicitly. Perceptual narrowing after 8–10 months (Werker and Tees, 1984) marks the point where stable predictions begin suppressing non-native contrasts.

Clinical dissociations sharpen the case. Developmental language disorder (DLD) impairs mass-count morphology but spares pragmatic article choice; autism spectrum disorder (ASD) shows the mirror profile (Creemers and Schaeffer, 2015). Eye-tracking confirms that DLD children still favour faces, where autistic peers don't (Vacas et al., 2024). Resting-state EEG and tractography add a matching double contrast (Campos et al., 2025; Li et al., 2024), and GWAS reports near-zero genetic correlation between DLD and ASD (Nudel et al., 2024). In this light, DLD looks like a domain-general auditory-motor constraint; ASD reflects atypical precision-weighting of social cues, the mechanism predicted here.

Adult L2 patterns and the DLD/ASD split together imply that attention, not processing capacity, is the principal bottleneck. Explicit instruction helps adults only when it shifts attention to low-salience features (Ellis, 2006). Minimal generic limits – oscillatory timing, shallow working memory – still bound learning, but only once precision sets the stage.

A parallel in AI underscores the point. Transformer models achieve state-of-the-art translation with attention alone, dispensing with recurrent or convolutional architectures once deemed essential (Vaswani et al., 2017). Biological and computational attention differ, yet both show that directing gain appropriately can replace specialized structure.

Predictive-processing theory casts this as adaptive priors that define what counts as surprising (Friston, Thornton, and Clark, 2012). Humans appear to inherit a prior that conspecific vocalizations are statistically rich. No hard-wired syntactic operator is needed; a bias to notice suffices. The bee waggle dance lay in plain sight until von Frisch decoded it (Frisch, 1967); without the right attentional filter, structure remains invisible.

3.5 Comparative implications: other vocal learners

If an attentional hyperprior is the pivotal human innovation, why do other vocal-learning taxa – songbirds, parrots, bats, cetaceans – fail to evolve language-like systems. The answer, on this account, is that they learn vocal patterns without the precision boost for hierarchical structure.

Songbirds illustrate the point. Zebra finches track species-typical frequency contours with millisecond accuracy but show scant evidence of attending to

supra-syllabic organization (Doupe and Kuhl, 1999). The grey parrot Alex, despite decades of model–rival training, required more than 30,000 hours to exceed one hundred referential labels, orders of magnitude slower than infant vocabulary growth (Pepperberg, 1999).

Marine mammals show the same asymmetry. Bottlenose dolphins can map acoustic tokens to objects or gestures, but gaining a few dozen symbols typically takes hundreds of reinforced trials per token and never yields spontaneous combinatorial use (Janik and P. J. B. Slater, 1997). The effort involved implies an attention system tuned to social calls, not to the nested regularities that scaffold human syntax.

Vocal learning, then, isn’t enough. The missing ingredient is a hyperprior that elevates the precision of ostensive, hierarchically organized signals, a feature documented in humans but, so far, in no other vocal learner.

3.6 Causal model

Figure 1 makes the identifying assumptions explicit. G represents genetic variation relevant to attention, C the quantity and quality of caregiver input, A the inherited attentional bias, L the learner’s internal trajectory (e.g. precision-weighted prediction-error dynamics), and Y the eventual language outcome. All genetic influence on language is assumed to flow through attention; no direct $G \rightarrow Y$ arrow is drawn. The only back-door from A to Y runs through C , so conditioning on caregiver input suffices to identify the causal effect $A \rightarrow Y$ (Pearl, 2009). (The proof is in the supplementary material.)

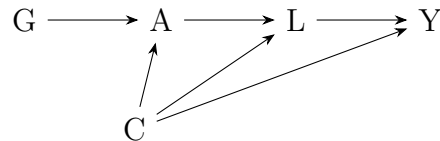


Figure 1: Core directed-acyclic graph for the attention-bias account. G , genetics; C , caregiver input; A , inherited attentional bias; L , learner trajectory (precision-weighted prediction-error dynamics); Y , language outcome. Controlling C blocks the sole back-door $A \leftarrow C \rightarrow Y$.

4 Testing the attention hypothesis

The hypothesis yields three falsifiable predictions: (i) an ostension-specific gain is present at birth in every human population; (ii) the gain is absent or strongly attenuated in other vocal learners; (iii) the gain alone can drive language-like learning in a generic neural architecture. Three complementary tests address these points.⁴

⁴Because caregiver input (C) is held constant while attention (A) is varied, the back-door $A \leftarrow C \rightarrow Y$ in Fig. 1 remains blocked.

4.1 Birth-onset cross-species preference test

Species: Human neonates (≤ 48 h), macaque neonates, and ferret kits (eyes open at birth, social, well-mapped auditory cortex).

Stimuli: Three amplitude- and spectrum-matched tracks filtered to each species' audiogram: infant-directed speech (IDS), modulation-matched rhythmic non-speech, and silence.

Measure: Pupil dilation and heart-rate deceleration, piloted in all taxa, with the less noisy metric retained.

Criterion: A human-only IDS preference before 48 h rules out postnatal learning and supports an inherited hyperprior.

Design: Sequential Bayes stopping ($\text{BF}_{10} > 10$ or < 0.1) minimises neonatal exposure. Parallel human cohorts (e.g., Toronto, Ho Chi Minh City) will check cross-cultural generality.

4.2 Computational replay with and without the gain

Capacity-matched neural models differ only in an attention scalar k : tokens in ostensive contexts receive loss $\times k$ (human-like), loss $\times 1$ (null), or loss scaled to environmental sounds (counterfactual). Multimodal corpora feed all models. Metrics: word-segmentation F1, subject-verb agreement accuracy, and hierarchical generalisation probes (Warstadt, Parrish, et al., 2020). A gain-only model that matches or exceeds human data while the null model fails would show that the attention tweak alone can bootstrap linguistic structure, satisfying prediction (iii).

4.3 Neonatal neural precision-weighting

Quiet-alert neonates (12–24 h) undergo fNIRS over bilateral superior temporal cortex. Conditions: IDS, adult-directed speech (ADS), and phase-scrambled speech matched for modulation spectrum. Target sample: 24 usable datasets (power for $d = .60$; 55 % attrition $\rightarrow$ 54 recruits).⁵ Precision metric: speech-to-cortex coherence at 4 Hz (Telkemeyer et al., 2009). Higher coherence IDS > ADS > scramble confirms an ostension-specific gain.

Together, the preference test (prediction i + ii), the in-silico replay (iii), and the neonatal fNIRS assay (i) offer a minimal yet decisive evaluation of the attention-bias account. Positive results would show that an inherited hyperprior for communicative signals, not specialized syntax circuitry, is the key evolutionary innovation behind human language.

⁵Anchored to the strongest infant effect in Lloyd-Fox et al. (2015): IDG-IDS > IDG-ADS, $t_{23} = 3.50$, $d_z \approx .71$. Mixed model: fixed CONDITION; random PARTICIPANT, OPTODE.

4.4 Does a modest precision boost matter? A back-of-the-envelope check

Take a toy learner that segments by comparing within-token transitional probabilities (TP_{in}) to cross-boundary probabilities (TP_{out}). Let half the tokens carry an ostensive flag and let those tokens receive a precision multiplier $g = 1.5$. Under a simple normal-approximation the standard error of the TP difference falls by the same factor; solving for the decision threshold θ shows that the sample size required for reliable boundary detection drops by about 25 % relative to uniform weighting (derivation in App. B). Even a 50 % gain boost delivers a sizeable bootstrap advantage under sparse input.

5 Implications for language emergence

An attentional hyperprior, not a bespoke syntax module, could be the sole genetic tweak behind language. Selection would have adjusted the gain on ostensive, hierarchically rhythmic input rather than inventing new processing machinery, a simpler trajectory within gene–culture coevolution.

That tweak dovetails with cultural transmission. Even weak learning biases, once amplified iteratively, yield systematic structure (Kirby, Cornish, and Smith, 2008). If the inherited bias is merely “listen harder to communicative rhythm,” culture supplies the rest.

Genetics offers no counter-evidence. Claims of a recent *FOXP2* sweep have collapsed under broader sampling (Atkinson et al., 2018). Genome-wide scans show diffuse polygenic signals (Hernandez et al., 2011; Mountford et al., 2022), consistent with selection acting on attention, motivation, and memory rather than on language-specific circuitry.

Comparative evidence supports this evolutionary trajectory. While nonhuman great ape infants receive minimal directed vocal input and have to rely on surrounding communication (Wegdell et al., 2025), human infants are immersed in ostensive signals from birth. This suggests that early hominins likely resembled other apes in their reliance on eavesdropping and observational learning, with the massive expansion of infant-directed communication co-evolving with language itself. The 400-fold increase in directed vocal input between humans and our closest relatives implicates a fundamental shift in attentional weighting rather than wholesale computational restructuring.

The proposal explains why language acquisition is universal while other sophisticated skills (e.g., echolocation, absolute pitch, formal mathematics) are rare. All humans share the attentional bias; songbirds, dolphins, and even highly enculturated primates don’t. The bias drives infants toward speech, then perceptual narrowing refines the mapping as experience accumulates (Werker and Tees, 1984). No extra genetic specification is required, matching Heyes’s broader claim that many human cognitive specialities are culturally scaffolded rather than hard-coded (Heyes, 2019).

6 Conclusion

Emergence, seen through an attentional lens, ceases to be a puzzle of hidden syntax machinery and becomes the predictable outcome of biased perception. A modest, inherited gain on ostensive, hierarchically rhythmic input can steer generic learning mechanisms toward grammar, leaving culture to supply the structural details. This single tweak aligns with gene–culture coevolution theory, bypasses the need for a language-specific module, and explains why language is universal while other complex skills aren’t.

The account yields clear, falsifiable predictions: an ostension bias at birth, its absence in non-human vocal learners, and its sufficiency in silico. Cross-cultural saccade-latency and ERP work, the neonatal preference and fNIRS studies outlined above, and attention-modulated neural models together form an empirical program that can confirm or refute the proposal.

Nothing here denies the intricacy of language; it relocates that intricacy to the interplay of a simple attentional prior with plastic neural circuitry and cultural iteration. We sought language in circuitry but find it in salience.

References

- Aihara, Ikkyu, Emyo Fujioka, and Shizuko Hiryu (2013). “Qualitative and quantitative analyses of the echolocation strategies of bats on the basis of mathematical modelling and laboratory experiments”. In: *PLoS ONE* 8.7, e68635. DOI: [10.1371/journal.pone.0068635](https://doi.org/10.1371/journal.pone.0068635).
- Atkinson, Elizabeth G. et al. (2018). “No Evidence for Recent Selection at FOXP2 among Diverse Human Populations”. In: *Cell* 174.6, 1424–1435.e15. DOI: [10.1016/j.cell.2018.06.048](https://doi.org/10.1016/j.cell.2018.06.048).
- Baker, C. L. (1979). “Syntactic theory and the projection problem”. In: *Linguistic Inquiry* 10.4, pp. 533–581.
- Beecher, Michael D., Inger M. Beecher, and Shari Hahn (1981). “Parent–offspring recognition in bank swallows (*Riparia riparia*): II. Development and acoustic basis”. In: *Animal Behaviour* 29.1, pp. 95–101. DOI: [10.1016/S0003-3472\(81\)80156-X](https://doi.org/10.1016/S0003-3472(81)80156-X).
- Behr, Oliver and Otto von Helversen (2004). “Bat serenades—complex courtship songs of the sac-winged bat (*Saccopteryx bilineata*)”. In: *Behavioral Ecology and Sociobiology* 56.2, pp. 106–115. DOI: [10.1007/s00265-004-0768-7](https://doi.org/10.1007/s00265-004-0768-7).
- Berwick, Robert C. and Noam Chomsky (2016). *Why Only Us: Language and Evolution*. Cambridge, MA: The MIT Press.
- Bérzin, F. and C. R. H. Fortinguerra (1993). “EMG study of the anterior, superior and posterior auricular muscles in man”. In: *Annals of Anatomy - Anatomischer Anzeiger* 175.2, pp. 195–197. DOI: [10.1016/S0940-9602\(11\)80182-2](https://doi.org/10.1016/S0940-9602(11)80182-2).
- Boë, Louis-Jean et al. (2019). “Which way to the dawn of speech?: Reanalyzing half a century of debates and data in light of speech science”. In: *Science Advances* 5.12, eaaw3916. DOI: [10.1126/sciadv.aaw3916](https://doi.org/10.1126/sciadv.aaw3916).
- Brumm, Henrik and Peter J.B. Slater (2006). “Animals can vary signal amplitude with receiver distance: Evidence from zebra finch song”. In: *Animal Behaviour* 72.3, pp. 699–705. DOI: [10.1016/j.anbehav.2006.01.020](https://doi.org/10.1016/j.anbehav.2006.01.020).
- Buchler, Edward R. (1980). “The development of flight, foraging, and echolocation in the little brown bat (*Myotis lucifugus*)”. In: *Behavioral Ecology and Sociobiology* 6.3, pp. 211–218. DOI: [10.1007/BF00569202](https://doi.org/10.1007/BF00569202).
- Burrows, Anne M. (2008). “The facial expression musculature in primates and its evolutionary significance”. In: *BioEssays* 30.3, pp. 212–225. DOI: [10.1002/bies.20719](https://doi.org/10.1002/bies.20719).
- Campos, Ana et al. (2025). “Resting-state electroencephalogram and speech perception in young children with developmental language disorder”. In: *Brain Sciences* 15.3, p. 219. DOI: [10.3390/brainsci15030219](https://doi.org/10.3390/brainsci15030219).
- Chomsky, Noam (1965). *Aspects of the Theory of Syntax*. Cambridge, MA: MIT Press.
- (1980). *Rules and Representations*. New York: Columbia University Press.
- Christiansen, Morten H and Nick Chater (2015). “The language faculty that wasn’t: A usage-based account of natural language recursion”. In: *Frontiers in Psychology* 6.AUG, pp. 1–18. DOI: [10.3389/fpsyg.2015.01182](https://doi.org/10.3389/fpsyg.2015.01182).

- Creemers, Ava and Jeannette C. Schaeffer (2015). “Grammatical and pragmatic properties of the DP in children with Specific Language Impairment and in children with High Functioning Autism”. In: *Linguistics in The Netherlands 2015*. John Benjamins, pp. 16–32.
- Dawkins, Richard (2024). *The Genetic Book of the Dead: A Darwinian Reverie*. New Haven: Yale University Press. DOI: [10.12987/9780300281668](https://doi.org/10.12987/9780300281668).
- Deacon, Terrence W (1997). *The Symbolic Species: The Co-evolution of Language and the Brain*. New York: W.W. Norton & Company.
- Dodsworth, C., L. J. Norman, and Lore Thaler (2020). “Navigation and perception of spatial layout in virtual echo-acoustic space”. In: *Cognition* 197, p. 104185. DOI: [10.1016/j.cognition.2020.104185](https://doi.org/10.1016/j.cognition.2020.104185).
- Doupe, Allison J. and Patricia K. Kuhl (1999). “Birdsong and human speech: Common themes and mechanisms”. In: *Annual Review of Neuroscience* 22, pp. 567–631. DOI: [10.1146/annurev.neuro.22.1.567](https://doi.org/10.1146/annurev.neuro.22.1.567).
- Ellis, Nick C (2006). “Selective attention and transfer phenomena in L2 acquisition: Contingency, cue competition, salience, interference, overshadowing, blocking, and perceptual learning”. In: *Applied Linguistics* 27.2, pp. 164–194.
- Evans, Nicholas and Stephen C. Levinson (2009). “The myth of language universals: Language diversity and its importance for cognitive science”. In: *Behavioral and Brain Sciences* 32.5, pp. 429–448. DOI: [10.1017/S0140525X0999094X](https://doi.org/10.1017/S0140525X0999094X).
- Farroni, Teresa et al. (2005). “Newborns’ preference for face-relevant stimuli: Effects of contrast polarity”. In: *Proceedings of the National Academy of Sciences of the United States of America* 102.47, pp. 17245–17250. DOI: [10.1073/pnas.0502205102](https://doi.org/10.1073/pnas.0502205102).
- Fitch, W. Tecumseh (2010). *The Evolution of Language*. Cambridge: Cambridge University Press.
- (2014). “Toward a computational framework for cognitive biology: Unifying approaches from cognitive neuroscience and comparative cognition”. In: *Physics of Life Reviews* 11.3, pp. 329–364. DOI: [10.1016/j.plrev.2014.04.005](https://doi.org/10.1016/j.plrev.2014.04.005).
- Fitch, W. Tecumseh, Marc D. Hauser, and Noam Chomsky (2005). “The evolution of the language faculty: Clarifications and implications”. In: *Cognition* 97.2, pp. 179–210. DOI: [10.1016/j.cognition.2005.02.005](https://doi.org/10.1016/j.cognition.2005.02.005).
- Fitch, W. Tecumseh and David Reby (2001). “The descended larynx is not uniquely human”. In: *Proceedings of the Royal Society of London. Series B: Biological Sciences* 268.1477, pp. 1669–1675. DOI: [10.1098/rspb.2001.1704](https://doi.org/10.1098/rspb.2001.1704).
- Fortin, Madeleine et al. (2008). “Wayfinding in the blind: Larger hippocampal volume and supranormal spatial navigation”. In: *Brain* 131.11, pp. 2995–3005. DOI: [10.1093/brain/awn250](https://doi.org/10.1093/brain/awn250).
- Friederici, Angela D. (2005). “Neurophysiological markers of early language acquisition: from syllables to sentences”. In: *Trends in Cognitive Sciences* 9.10, pp. 481–488. DOI: [10.1016/j.tics.2005.08.008](https://doi.org/10.1016/j.tics.2005.08.008).
- Frisch, Karl von (1967). *The dance language and orientation of bees*. Trans. by Leigh E. Chadwick. Cambridge, Mass: The Belknap Press.

- Friston, Karl J., Christopher Thornton, and Andy Clark (2012). “Free-energy minimization and the dark-room problem”. In: *Frontiers in Psychology* 3. DOI: [10.3389/fpsyg.2012.00130](https://doi.org/10.3389/fpsyg.2012.00130).
- Gould, Edwin, Norman C. Negus, and Alvin Novick (1964). “Evidence for echolocation in shrews”. In: *Journal of Experimental Zoology* 156.1, pp. 19–37. DOI: [10.1002/jez.1401560103](https://doi.org/10.1002/jez.1401560103).
- Gould, Stephen J. and Richard C. Lewontin (1979). “The spandrels of San Marco and the Panglossian paradigm: A critique of the adaptationist programme”. In: *Proceedings of the Royal Society of London. Series B, Biological Sciences* 205.1161, pp. 581–598. DOI: [10.1098/rspb.1979.0086](https://doi.org/10.1098/rspb.1979.0086).
- Gould, Stephen Jay and Elisabeth S. Vrba (1982). “Exaptation—A missing term in the science of form”. In: *Paleobiology* 8.1, pp. 4–15. DOI: [10.1017/S0094837300004310](https://doi.org/10.1017/S0094837300004310).
- Hare, Brian et al. (2005). “Social cognitive evolution in captive foxes is a correlated by-product of experimental domestication”. In: *Current Biology* 15.3, pp. 226–230. DOI: [10.1016/j.cub.2005.01.040](https://doi.org/10.1016/j.cub.2005.01.040).
- Hauser, Marc D., Noam Chomsky, and W. Tecumseh Fitch (2002). “The faculty of language: What is it, who has it, and how did it evolve?” In: *Science* 298.5598, pp. 1569–1579. DOI: [10.1126/science.298.5598.1569](https://doi.org/10.1126/science.298.5598.1569).
- Hernandez, Ryan D. et al. (2011). “Classic selective sweeps were rare in recent human evolution”. In: *Science* 331.6019, pp. 920–924. DOI: [10.1126/science.1198878](https://doi.org/10.1126/science.1198878).
- Heyes, Cecilia (2019). “Précis of *Cognitive gadgets: The cultural evolution of thinking*”. In: *Behavioral and Brain Sciences* 42, e169. DOI: [10.1017/S0140525X18002145](https://doi.org/10.1017/S0140525X18002145).
- Hutcheon, James M., John A. W. Kirsch, and Jr. Garland Theodore (2002). “A comparative analysis of brain size in relation to foraging ecology and phylogeny in the Chiroptera”. In: *Brain, Behavior and Evolution* 60.3, pp. 165–180. DOI: [10.1159/000065938](https://doi.org/10.1159/000065938).
- Immelmann, Klaus (1969). “Song development in the zebra finch and other estrildid finches”. In: *Bird Vocalizations*. Ed. by Robert A. Hinde. Cambridge University Press, pp. 61–74. URL: <https://api.semanticscholar.org/CorpusID:86502555>.
- Jackendoff, Ray (2002). *Foundations of Language: Brain, Meaning, Grammar, Evolution*. Oxford: Oxford University Press. DOI: [10.1093/acprof:oso/9780198270126.001.0001](https://doi.org/10.1093/acprof:oso/9780198270126.001.0001).
- Janik, Vincent M. and Peter J. B. Slater (1997). “Vocal learning in mammals”. In: *Advances in the Study of Behavior* 26, pp. 59–99. DOI: [10.1016/S0065-3454\(08\)60377-0](https://doi.org/10.1016/S0065-3454(08)60377-0).
- (1998). “Context-specific use suggests that bottlenose dolphin signature whistles are cohesion calls”. In: *Animal Behaviour* 56, pp. 829–838. DOI: [10.1006/anbe.1998.0881](https://doi.org/10.1006/anbe.1998.0881).
- Johnson, Mark H. et al. (1991). “Newborns’ preferential tracking of face-like stimuli and its subsequent decline”. In: *Cognition* 40.1–2, pp. 1–19. DOI: [10.1016/0010-0277\(91\)90045-6](https://doi.org/10.1016/0010-0277(91)90045-6).

- Jordan, Gabriele et al. (2010). “The dimensionality of color vision in carriers of anomalous trichromacy”. In: *Journal of Vision* 10.8. DOI: [10.1167/10.8.12](https://doi.org/10.1167/10.8.12).
- Kallini, Julie et al. (2024). “Mission: Impossible Language Models”. In: arXiv: [2401.06416 \[cs.CL\]](https://arxiv.org/abs/2401.06416). URL: <https://arxiv.org/abs/2401.06416>.
- King, Stephanie L. and Vincent M. Janik (2013). “Bottlenose dolphins can use learned vocal labels to address each other”. In: *Proceedings of the National Academy of Sciences* 110, pp. 13216–13221. DOI: [10.1073/pnas.1304459110](https://doi.org/10.1073/pnas.1304459110).
- Kirby, Simon, Hannah Cornish, and Kenny Smith (2008). “Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language”. In: *Proceedings of the National Academy of Sciences of the United States of America* 105.31, pp. 10681–10686. DOI: [10.1073/pnas.0707835105](https://doi.org/10.1073/pnas.0707835105).
- Knörnschild, Mirjam et al. (2010). “Complex vocal imitation during ontogeny in a bat”. In: *Biology Letters* 6.2, pp. 156–159. DOI: [10.1098/rsbl.2009.0685](https://doi.org/10.1098/rsbl.2009.0685).
- Kolarik, Andrew J. et al. (2014). “A summary of research investigating echolocation abilities of blind and sighted humans”. In: *Hearing Research* 310, pp. 60–68. DOI: [10.1016/j.heares.2014.01.010](https://doi.org/10.1016/j.heares.2014.01.010).
- Kolodny, Oren and Shimon Edelman (2018). “The evolution of the capacity for language: The ecological context and adaptive value of a process of cognitive hijacking”. In: *Philosophical Transactions of the Royal Society B* 373.1743, p. 20170052. DOI: [10.1098/rstb.2017.0052](https://doi.org/10.1098/rstb.2017.0052).
- Kuhl, Patricia K. (2004). “Early language acquisition: Cracking the speech code”. In: *Nature Reviews Neuroscience* 5, pp. 831–843. DOI: [10.1038/nrn1533](https://doi.org/10.1038/nrn1533).
- Kuhl, Patricia K., Feng-Ming Tsao, and Huei-Mei Liu (2003). “Foreign-Language Experience in Infancy: Effects of Short-Term Exposure and Social Interaction on Phonetic Learning”. In: *Proceedings of the National Academy of Sciences* 100.15, pp. 9096–9101. DOI: [10.1073/pnas.1532872100](https://doi.org/10.1073/pnas.1532872100).
- Large, Edward W. and Caroline Palmer (2002). “Perceiving temporal regularity in music”. In: *Cognitive Science* 26, pp. 1–37. DOI: [10.1016/S0364-0213\(01\)00057-X](https://doi.org/10.1016/S0364-0213(01)00057-X).
- Li, Min et al. (2024). “Altered white matter connectivity of ventral language networks in autism spectrum disorder: An automated fiber quantification analysis with multi-site datasets”. In: *NeuroImage* 297, p. 120731. DOI: [10.1016/j.neuroimage.2024.120731](https://doi.org/10.1016/j.neuroimage.2024.120731).
- Lieberman, Philip (2017). “Comment on “Monkey vocal tracts are speech-ready””. In: *Science Advances* 3.7, e1700442. DOI: [10.1126/sciadv.1700442](https://doi.org/10.1126/sciadv.1700442).
- Lloyd-Fox, Sarah et al. (2015). “Are you talking to me? Neural activations in 6-month-old infants in response to being addressed during natural interactions”. In: *Cortex* 70, pp. 35–48. DOI: [10.1016/j.cortex.2015.02.005](https://doi.org/10.1016/j.cortex.2015.02.005).
- Milne, Jennifer L. et al. (2015). “Parahippocampal cortex is involved in material processing via echoes in blind echolocation experts”. In: *Vision Research* 109, pp. 139–148. DOI: [10.1016/j.visres.2014.07.004](https://doi.org/10.1016/j.visres.2014.07.004).
- Mountford, Hayley S. et al. (2022). “The Genetic and Molecular Basis of Developmental Language Disorder: A Review”. In: *Children* 9.5, p. 586. DOI: [10.3390/children9050586](https://doi.org/10.3390/children9050586).

- Nakahara, Fumio and Nobuyuki Miyazaki (2011). “Vocal exchanges of signature whistles in bottlenose dolphins (*Tursiops truncatus*)”. In: *Journal of Ethology* 29.2, pp. 309–320. DOI: [10.1007/s10164-010-0259-4](https://doi.org/10.1007/s10164-010-0259-4).
- Norman, Liam J., Tom Hartley, and Lore Thaler (2024). “Changes in primary visual and auditory cortex of blind and sighted adults following 10 weeks of click-based echolocation training”. In: *Cerebral Cortex* 34.6, bhae239. DOI: [10.1093/cercor/bhae239](https://doi.org/10.1093/cercor/bhae239).
- Norman, Liam J. and Lore Thaler (2019). “Retinotopic-like maps of spatial sound in primary ‘visual’ cortex of blind human echolocators”. In: *Proceedings of the Royal Society B: Biological Sciences* 286.1912, p. 20191910. DOI: [10.1098/rspb.2019.1910](https://doi.org/10.1098/rspb.2019.1910).
- (2023). “The Occipital Place Area is Recruited for Echo-Acoustically Guided Navigation in Blind Human Echolocators”. In: *Journal of Neuroscience* 43.24, pp. 4470–4486. DOI: [10.1523/JNEUROSCI.1402-22.2023](https://doi.org/10.1523/JNEUROSCI.1402-22.2023).
- Nudel, Ron et al. (2024). “Developmental language disorder – heritability and genetic correlations with other disorders affecting language”. In: *Psychiatry Research* 342, p. 116212. DOI: [10.1016/j.psychres.2024.116212](https://doi.org/10.1016/j.psychres.2024.116212).
- Ocklenburg, Sebastian, Felix Ströckens, and Onur Güntürkün (2013). “Lateralisation of conspecific vocalisation in non-human vertebrates”. In: *Laterality: Asymmetries of Body, Brain and Cognition* 18.1, pp. 1–31. DOI: [10.1080/1357650X.2011.626561](https://doi.org/10.1080/1357650X.2011.626561).
- Patel, Aniruddh D. et al. (2009). “Experimental evidence for synchronization to a musical beat in a nonhuman animal”. In: *Current Biology* 19, pp. 827–830. DOI: [10.1016/j.cub.2009.03.038](https://doi.org/10.1016/j.cub.2009.03.038).
- Pearl, Judea (2009). *Causality: Models, Reasoning, and Inference*. 2nd ed. Cambridge: Cambridge University Press.
- Pepperberg, Irene Maxine (1999). *The Alex Studies: Cognitive and Communicative Abilities of Grey Parrots*. Cambridge, MA: Harvard University Press. DOI: [10.4159/9780674041998](https://doi.org/10.4159/9780674041998).
- Perfors, Andrew, Joshua B. Tenenbaum, and Terry Regier (2011). “The Learnability of Abstract Syntactic Principles”. In: *Cognition* 118.3, pp. 306–338. DOI: [10.1016/j.cognition.2010.11.001](https://doi.org/10.1016/j.cognition.2010.11.001).
- Pinker, Steven and Paul Bloom (1990). “Natural language and natural selection”. In: *Behavioral and Brain Sciences* 13.4, pp. 707–784. DOI: [10.1017/S0140525X00081061](https://doi.org/10.1017/S0140525X00081061).
- Pinker, Steven and Ray Jackendoff (2005). “The faculty of language: What’s special about it?” In: *Cognition* 95.2, pp. 201–236. DOI: [10.1016/j.cognition.2004.08.004](https://doi.org/10.1016/j.cognition.2004.08.004).
- Putnam, Hilary (1980). “What is Innate and Why”. In: *Language and Learning: The Debate Between Jean Piaget and Noam Chomsky*. Ed. by Massimo Piattelli-Palmarini. Harvard University Press.
- Rademacher, J. et al. (2001). “Probabilistic mapping and volume measurement of human primary auditory cortex”. In: *NeuroImage* 13.4, pp. 669–683. DOI: [10.1006/nimg.2000.0714](https://doi.org/10.1006/nimg.2000.0714).

- Reynolds, Brett (2020). *What echolocation does and doesn't mean for language*. OSF preprint. DOI: [10.31234/osf.io/h7qdr](https://doi.org/10.31234/osf.io/h7qdr).
- Rokicki, Jaroslav et al. (2022). "Oxytocin receptor expression patterns in the human brain across development". In: *Neuropsychopharmacology* 47.8, pp. 1550–1560. DOI: [10.1038/s41386-022-01305-5](https://doi.org/10.1038/s41386-022-01305-5).
- Rombaux, Philippe et al. (2010). "Increased olfactory bulb volume and olfactory function in early blind subjects". In: *NeuroReport* 21.17, pp. 1069–1073. DOI: [10.1097/WNR.0b013e32833fcb8a](https://doi.org/10.1097/WNR.0b013e32833fcb8a).
- Rosenzweig, Mark R., Donald A. Riley, and David Krech (1955). "Evidence for echolocation in the rat". In: *Science* 121.3147, p. 600. DOI: [10.1126/science.121.3147.600](https://doi.org/10.1126/science.121.3147.600).
- Saffran, Jenny R., Richard N. Aslin, and Elissa L. Newport (1996). "Statistical Learning by 8-Month-Old Infants". In: *Science* 274.5294, pp. 1926–1928. DOI: [10.1126/science.274.5294.1926](https://doi.org/10.1126/science.274.5294.1926).
- Schmidt, Richard W (1990). "The role of consciousness in second language learning". In: *Applied Linguistics* 11.2, pp. 129–158.
- Scholz, Barbara C. et al. (2025). "Philosophy of Linguistics". In: *The Stanford Encyclopedia of Philosophy*. Ed. by Edward N. Zalta and Uri Nodelman. Summer 2025. Metaphysics Research Lab, Stanford University.
- Scott-Phillips, Thom (2015). *Speaking Our Minds: Why Human Communication Is Different, and How Language Evolved to Make It Special*. Basingstoke: Palgrave Macmillan.
- (2017). "Pragmatics and the Aims of Language Evolution". In: *Psychonomic Bulletin & Review* 24.1, pp. 186–189. DOI: [10.3758/s13423-016-1061-2](https://doi.org/10.3758/s13423-016-1061-2).
- Seligman, Martin E. P. (1970). "On the generality of the laws of learning". In: *Psychological Review* 77.5, pp. 406–418. DOI: [10.1037/h0029790](https://doi.org/10.1037/h0029790).
- Shibasaki, Masahiro and Nobuyuki Kawai (2009). "Rapid detection of snakes by Japanese monkeys (*Macaca fuscata*): an evolutionarily predisposed visual system". In: *Journal of Comparative Psychology* 123.2, pp. 131–135. DOI: [10.1037/a0015095](https://doi.org/10.1037/a0015095).
- Telkemeyer, Silke et al. (2009). "Sensitivity of Newborn Auditory Cortex to the Temporal Structure of Sounds". In: *Journal of Neuroscience* 29.47, pp. 14726–14733. DOI: [10.1523/JNEUROSCI.1246-09.2009](https://doi.org/10.1523/JNEUROSCI.1246-09.2009).
- Teng, Santani and David Whitney (2011). "The acuity of echolocation: Spatial resolution in the sighted compared to expert performance". In: *Journal of Visual Impairment & Blindness* 105.1, pp. 20–32. DOI: [10.1177/0145482X1110500103](https://doi.org/10.1177/0145482X1110500103).
- Terrace, Herbert S. (1979). *Nim: A Chimpanzee Who Learned Sign Language*. New York: Knopf.
- Thaler, Lore (2013). "Echolocation may have real-life advantages for blind people: An analysis of survey data". In: *Frontiers in Physiology* 4. DOI: [10.3389/fphys.2013.00098](https://doi.org/10.3389/fphys.2013.00098).
- Thompson, Bill, Simon Kirby, and Kenny Smith (2016). "Culture shapes the evolution of cognition". In: *Proceedings of the National Academy of Sciences USA* 113.16, pp. 4530–4535. DOI: [10.1073/pnas.1523631113](https://doi.org/10.1073/pnas.1523631113).

- Tomasello, Michael (2003). *Constructing a Language: A Usage-Based Theory of Language Acquisition*. Cambridge, MA: Harvard University Press. DOI: [10.2307/j.ctv26070v8](https://doi.org/10.2307/j.ctv26070v8).
- Tooby, John, Leda Cosmides, and H. Clark Barrett (2005). “Resolving the debate on innate ideas: Learnability constraints and the evolved interpenetration of motivational and conceptual functions”. In: *The innate mind: Structure and contents*. Ed. by Peter Carruthers, Stephen Laurence, and Stephen Stich. Oxford: Oxford University Press, pp. 305–337. DOI: [10.1093/acprof:oso/9780195179675.003.0018](https://doi.org/10.1093/acprof:oso/9780195179675.003.0018).
- Vacas, Julia et al. (2024). “Eye-Tracking Methodology to Detect Differences in Attention to Faces Between Developmental Language Disorder and Autism”. In: *Journal of Speech, Language, and Hearing Research* 67.9, pp. 3148–3162. DOI: [10.1044/2024_JSLHR-24-00109](https://doi.org/10.1044/2024_JSLHR-24-00109).
- van der Aa, Jeroen, Henkjan Honing, and Carel ten Cate (2015). “The perception of regularity in an isochronous stimulus in zebra finches (*Taeniopygia guttata*) and humans”. In: *Behavioural Processes* 115, pp. 37–45. DOI: [10.1016/j.beproc.2015.02.018](https://doi.org/10.1016/j.beproc.2015.02.018).
- Vaswani, Ashish et al. (2017). “Attention is all you need”. In: *Advances in Neural Information Processing Systems* 30, pp. 5998–6008.
- Vater, Marianne, Jin Meng, and Richard C. Fox (2004). “Hearing Organ Evolution and Specialization: Early and Later Mammals”. In: *Evolution of the Vertebrate Auditory System*. Ed. by Geoffrey A. Manley, Richard R. Fay, and Arthur N. Popper. New York, NY: Springer New York, pp. 256–288. DOI: [10.1007/978-1-4419-8957-4_9](https://doi.org/10.1007/978-1-4419-8957-4_9).
- Vernes, Sonja C. (2017). “What bats have to say about speech and language”. In: *Psychonomic Bulletin & Review* 24.1, pp. 111–117. DOI: [10.3758/s13423-016-1060-3](https://doi.org/10.3758/s13423-016-1060-3).
- Volman, Susan F. (1993). “Development of Neural Selectivity for Birdsong during Vocal Learning”. In: *Journal of Neuroscience* 13.11, pp. 4737–4747. DOI: [10.1523/JNEUROSCI.13-11-04737.1993](https://doi.org/10.1523/JNEUROSCI.13-11-04737.1993).
- Vouloumanos, Athena and Janet F. Werker (2007). “Listening to language at birth: Evidence for a bias for speech in neonates”. In: *Developmental Science* 10.2, pp. 159–164. DOI: [10.1111/j.1467-7687.2007.00549.x](https://doi.org/10.1111/j.1467-7687.2007.00549.x).
- Warstadt, Alex, Aaron Mueller, et al. (2023). “Findings of the BabyLM Challenge: Sample-Efficient Pretraining on Developmentally Plausible Corpora”. In: *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning*. Singapore: Association for Computational Linguistics, pp. 1–34. DOI: [10.18653/v1/2023.conll-babylm.1](https://doi.org/10.18653/v1/2023.conll-babylm.1).
- Warstadt, Alex, Alicia Parrish, et al. (2020). “BLiMP: The Benchmark of Linguistic Minimal Pairs for English”. In: *Transactions of the Association for Computational Linguistics* 8, pp. 377–392. DOI: [10.1162/tac1_a_00321](https://doi.org/10.1162/tac1_a_00321).
- Wegdell, Franziska et al. (2025). “The evolution of infant-directed communication: Comparing vocal input across all great apes”. In: *Science Advances* 11.26. DOI: [10.1126/sciadv.adt7718](https://doi.org/10.1126/sciadv.adt7718).

- Werker, Janet F. and Richard C. Tees (1984). “Cross-language speech perception: Evidence for perceptual reorganization during the first year of life”. In: *Infant Behavior and Development* 7.1, pp. 49–63. ISSN: 0163-6383. DOI: [10.1016/S0163-6383\(84\)80022-3](https://doi.org/10.1016/S0163-6383(84)80022-3).
- Wilkins, John F., Richard W. Wrangham, and W. Tecumseh Fitch (2014). “The “Domestication Syndrome” in Mammals: A Unified Explanation Based on Neural Crest Cell Behavior and Genetics”. In: *Genetics* 197.3, pp. 795–808. DOI: [10.1534/genetics.114.165423](https://doi.org/10.1534/genetics.114.165423).
- Wilkinson, Gerald S. (2003). “Social and vocal complexity in bats”. In: *Animal social complexity: Intelligence, culture, and individualized societies*. Ed. by Frans B. M. de Waal and Peter L. Tyack. Cambridge, MA: Harvard University Press, pp. 322–341. DOI: [10.4159/harvard.9780674419131.c24](https://doi.org/10.4159/harvard.9780674419131.c24).
- Wohlers, David W. and Franz Huber (1982). “Processing of Sound Signals by Six Types of Neurons in the Prothoracic Ganglion of the Cricket, *Gryllus campestris* L.” In: *Journal of Comparative Physiology A* 146.2, pp. 161–173. DOI: [10.1007/BF00610234](https://doi.org/10.1007/BF00610234).
- Wrangham, Richard W. and Brian Hare (2019). “Hypotheses for the Evolution of Reduced Reactive Aggression in the Self-Domesticated Hominins”. In: *Frontiers in Psychology* 10, p. 1914. DOI: [10.3389/fpsyg.2019.01914](https://doi.org/10.3389/fpsyg.2019.01914).
- Zanella, Matteo et al. (2019). “Dosage analysis of the 7q11.23 Williams region identifies *BAZ1B* as a major human gene patterning the modern human face and underlying self-domestication”. In: *Science Advances* 5.12, eaaw7908. DOI: [10.1126/sciadv.aaw7908](https://doi.org/10.1126/sciadv.aaw7908).